Unit 6

Moving from an Idea to an Entrepreneurial Firm

3 Slibraries · 15 framework sheets

Blank's 4 Steps

SLIBRARY 16 · SHEET 1 / 5

Discovery · Validation · Creation · Company Building

PROFESSIONAL DEFINITION

Steve Blank's Customer Development methodology decomposes the venture journey into four sequenced steps: *Customer Discovery* (problem-solution fit), *Customer Validation* (repeatable sales process), *Customer Creation* (scaling demand), and *Company Building* (transitioning from startup to operating company). Each step has its own metrics, gates and failure modes; skipping or compressing the sequence is the dominant cause of well-funded startup failure (Blank, 2005, 2013).

WHO DEVELOPED IT

Steve Blank, eight-time founder and Stanford/Berkeley/Columbia adjunct professor, in *The Four Steps to the Epiphany* (2005). Refined for HBR audiences in his 2013 article "Why the Lean Start-Up Changes Everything" and elaborated in *The Startup Owner's Manual* (Blank and Dorf, 2012).

WHY IT WAS DEVELOPED

Blank observed that successful startups followed a different sequence than the product-development methodology mainstream business education taught. The product-development model assumed customers were known; Blank's customer-development model treated customer discovery as a parallel and prior process.

FIGURE 1 · FOUR STAGES

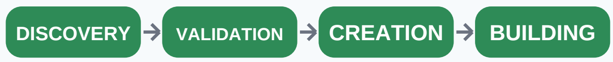

Figure 1. Each stage has its own gate; skipping or compressing the sequence is the dominant failure mode. Adapted from Blank (2005).

HOW IT IS USED

Walk the four steps in sequence with explicit gates between them. Discovery: confirm problem and solution. Validation: confirm repeatable sales/usage. Creation: scale acquisition. Building: institutionalise as an operating company. Pivot when a step's gate cannot be passed.

WHEN IT IS USED

Use as the venture-stage roadmap for any startup or corporate-innovation venture, in accelerator-programme curriculum design, and in venture-capital staged-funding decisions.

BY WHOM IT IS USED

Founders, accelerator coaches, venture-capital partners, MBA entrepreneurship instructors, and corporate-innovation programme directors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Cooper's (1990) stage-gate process was designed for established firms with known customers, Blank's contribution is the customer-development inversion — treating customer discovery as a primary, parallel process. Building on Crawford's (1994) NPD literature, it gave Lean Startup its sequenced architecture.

RELATED FRAMEWORKS IN THIS COURSE

- Discovery Interviews · Slibrary 16
- Validation Experiments · Slibrary 16
- Problem-Solution Fit · Slibrary 14
- Product-Market Fit · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Blank, S. (2005) *The Four Steps to the Epiphany*. Pescadero: K&S; Ranch.
Blank, S. (2013) 'Why the Lean Start-Up Changes Everything', *Harvard Business Review*, 91(5), pp. 63–72.
Blank, S. and Dorf, B. (2012) *The Startup Owner's Manual*. Pescadero: K&S; Ranch.
Cooper, R.G. (1990) 'Stage-Gate Systems', *Business Horizons*, 33(3), pp. 44–54.

Discovery Interviews

SLIBRARY 16 · SHEET 2 / 5

20–30 customer interviews before you write a line of code

PROFESSIONAL DEFINITION

Discovery Interviews are structured, qualitative customer conversations conducted before product build to validate problem existence and severity. The discipline requires 20–30 interviews per segment, scripted to elicit recent specific behaviour rather than hypothetical preferences, and analysed for patterns that reveal under-served jobs. The interviews are the most cost-effective evidence-gathering method in early-stage venture work — the best 4 weeks any pre-revenue founder can spend (Blank and Dorf, 2012; Constable, 2014).

WHO DEVELOPED IT

Steve Blank in *The Four Steps to the Epiphany* (2005) and operationalised by Giff Constable in *Talking to Humans* (2014). Roots in Beyer and Holtzblatt's (1998) contextual-inquiry tradition and Cooper's (1999) persona-research methodology.

WHY IT WAS DEVELOPED

Founders skipped customer research because they "knew" the customer (often without ever having spoken to one). Blank made discovery interviews the non-negotiable first step of customer development; Constable's tactical guide removed the excuse of not knowing how.

FIGURE 1 · INTERVIEW DISCIPLINE

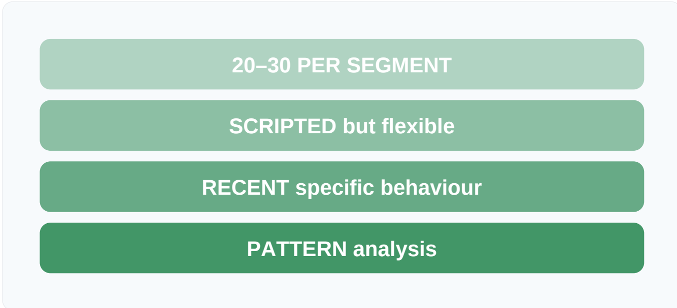

Figure 1. Single interviews are anecdotes; patterns across 20–30 are evidence. Adapted from Constable (2014).

HOW IT IS USED

Recruit 20–30 representative target customers. Use a scripted but flexible protocol: open with their context, dig into recent specific behaviour (not preferences), surface jobs and pains. Analyse for patterns across interviews — single-interview findings are anecdotes.

WHEN IT IS USED

Use as the first work of any new venture, before MVP build, before any product-roadmap commitment, and in pivot decisions when team conviction has outrun evidence.

BY WHOM IT IS USED

Founders, product researchers, accelerator coaches running structured discovery cohorts, and corporate-innovation labs validating internal venture proposals.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Cooper's (1999) personas captured user attributes and Beyer and Holtzblatt's (1998) contextual inquiry observed work-in-context, Constable's contribution is the venture-stage tactical playbook — script, recruitment, pattern-analysis. Building on Blank's (2005) sequence, it makes discovery interviewing teachable to non-researchers.

RELATED FRAMEWORKS IN THIS COURSE

- Blank's 4 Steps · Slibrary 16
- Switch Interview · 4 Forces · Slibrary 13
- Validation Experiments · Slibrary 16
- JTBD Canvas · Slibrary 13

SOURCES (HARVARD REFERENCING STYLE)

Constable, G. (2014) *Talking to Humans*. Self-published.
Blank, S. and Dorf, B. (2012) *The Startup Owner's Manual*. Pescadero: K&S; Ranch.
Blank, S. (2005) *The Four Steps to the Epiphany*. Pescadero: K&S; Ranch.
Beyer, H. and Holtzblatt, K. (1998) *Contextual Design*. San Francisco: Morgan Kaufmann.

Validation Experiments

SLIBRARY 16 · SHEET 3 / 5

Pre-order · Concierge · Fake-door · Smoke-test — patterns that prove demand

PROFESSIONAL DEFINITION

Validation Experiments are structured tests that move beyond stated preference to elicit real customer behaviour: *Pre-order* (do they pay before delivery?), *Concierge* (does manual delivery satisfy them?), *Fake-door* (do they click through to the offer?), *Smoke-test* (does the landing page convert?). Each pattern produces falsifiable evidence about a specific risky assumption — the only evidence that should justify build investment (Bland and Osterwalder, 2019; Maurya, 2012).

WHO DEVELOPED IT

Codified by David Bland and Alexander Osterwalder in *Testing Business Ideas* (2019), drawing on Lean Startup practices documented by Ries (2011) and Maurya (2012). Each individual pattern has its own genealogy — fake-door tests trace to Yelp's early traction tests, smoke-tests to Buffer's launch.

WHY IT WAS DEVELOPED

Customer interviews surfaced stated preferences that were unreliable predictors of actual purchase. Validation experiments substituted real-money or real-clicks as evidence, replacing stated-preference data with revealed-preference data — the gold standard of demand validation.

FIGURE 1 · FOUR PATTERNS

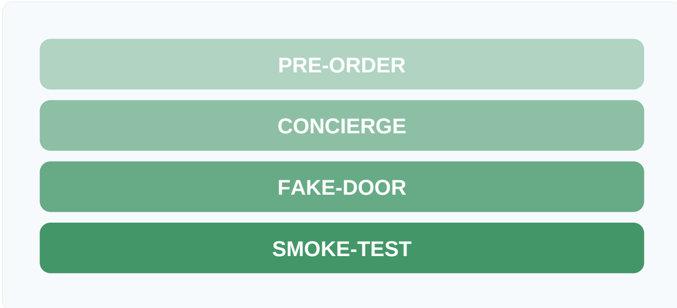

Figure 1. Each pattern produces revealed-preference evidence on a specific assumption type.

HOW IT IS USED

Match the experiment pattern to the assumption being tested. Pre-orders test willingness to pay. Concierge tests value-delivery feasibility. Fake-doors test acquisition-channel demand. Smoke-tests test landing-page conversion. Set explicit success thresholds before running.

WHEN IT IS USED

Use after discovery interviews when patterns suggest a hypothesis worth testing, before MVP build, in stage-gate validation reviews, and in venture-capital due diligence on portfolio companies.

BY WHOM IT IS USED

Founders, lean-startup coaches, growth marketers, accelerator coaches, and corporate-innovation labs validating internal venture proposals.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Ries's (2011) MVP was a single class of experiment and Maurya's (2012) Lean Canvas catalogued risks, Bland and Osterwalder's contribution is the comprehensive 44-pattern test library mapping assumption types to specific experiments. Building on Christensen's (2003) discovery-driven planning, it makes test design systematic.

RELATED FRAMEWORKS IN THIS COURSE

- Riskiest Assumption Test · Slibrary 15
- Discovery Interviews · Slibrary 16
- Pretotyping · Slibrary 15
- Build-Measure-Learn · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

Bland, D.J. and Osterwalder, A. (2019) *Testing Business Ideas*. Hoboken: Wiley.
Maurya, A. (2012) *Running Lean*. 2nd edn. Sebastopol: O'Reilly.
Ries, E. (2011) *The Lean Startup*. New York: Crown Business.
Christensen, C.M. and Raynor, M.E. (2003) *The Innovator's Solution*. Boston: Harvard Business School Press.

AARRR · Pirate Metrics

SLIBRARY 16 · SHEET 4 / 5

Acquisition · Activation · Retention · Referral · Revenue

PROFESSIONAL DEFINITION

Dave McClure's AARRR ("Pirate Metrics") frames startup growth as a five-stage funnel: *Acquisition* (visitors arrive), *Activation* (first valuable experience), *Retention* (repeat use), *Referral* (existing users invite new ones), and *Revenue* (monetisation). Each stage has its own metric and conversion rate; the bottleneck stage (lowest conversion) is the highest-leverage growth target. The framework concentrates growth investment on the actual bottleneck rather than the easiest stage to instrument (McClure, 2007).

WHO DEVELOPED IT

Dave McClure, 500 Startups founder, in his 2007 conference talk and blog posts subsequently formalised in startup-growth literature. The framework synthesises classical marketing-funnel concepts with behavioural-cohort analysis emerging from Web 2.0 product-analytics practice.

WHY IT WAS DEVELOPED

Startups optimised whichever funnel stage they could measure most easily — usually acquisition (web traffic) — while ignoring stages where actual value was created or destroyed (activation, retention). McClure's framework forced explicit measurement of all five and concentrated investment on the actual bottleneck.

FIGURE 1 · FIVE-STAGE FUNNEL

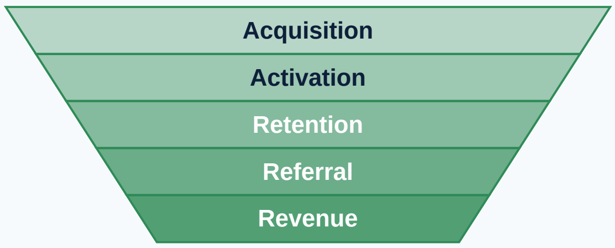

Figure 1. The bottleneck stage is the highest-leverage growth target. Adapted from McClure (2007).

HOW IT IS USED

Instrument all five stages with cohort-based metrics. Identify the bottleneck (lowest conversion stage). Concentrate growth experiments on the bottleneck until it is no longer the constraint, then re-identify the new bottleneck.

WHEN IT IS USED

Use as the growth-strategy operating frame from product-launch onward, in growth-team weekly reviews, in venture-capital growth-stage portfolio diagnostics, and in product-analytics implementation.

BY WHOM IT IS USED

Growth marketers, founders, product analytics teams, and venture-capital partners diagnosing portfolio-company growth health.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Lewis's (1898) AIDA funnel was a marketing-attention model and traditional sales funnels (Strong, 1925) measured deal stages, McClure's contribution is the stage-and-cohort-aware product-engagement funnel for digital ventures. Building on Goldratt's (1984) theory of constraints, it concentrates growth investment on the bottleneck.

RELATED FRAMEWORKS IN THIS COURSE

Traction Canvas · Slibrary 14
Customer Journey Map · Slibrary 3
Product-Market Fit · Slibrary 14
Lean Stack · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

McClure, D. (2007) 'Startup Metrics for Pirates: AARRR!', *500 Startups blog*.
Croll, A. and Yoskovitz, B. (2013) *Lean Analytics: Use Data to Build a Better Startup Faster*. Sebastopol: O'Reilly.
Skok, D. (2017) 'SaaS Metrics 2.0', *For Entrepreneurs blog*.
Goldratt, E.M. (1984) *The Goal: A Process of Ongoing Improvement*. Great Barrington: North River Press.

Pivot Types

SLIBRARY 16 · SHEET 5 / 5

Zoom-in · Zoom-out · Customer · Need · Platform · Tech · Channel · Revenue · Engine

PROFESSIONAL DEFINITION

Eric Ries's pivot taxonomy classifies the directions a startup can pivot when build-measure-learn evidence indicates the current strategy is failing: *Zoom-in* (a feature becomes the product), *Zoom-out* (the product becomes a feature), *Customer-segment*, *Customer-need*, *Platform*, *Business-architecture*, *Value-capture*, *Engine-of-growth*, *Channel*, and *Technology* pivots. The taxonomy converts "we need to pivot" from a vague signal into a specific decision (Ries, 2011).

WHO DEVELOPED IT

Eric Ries in *The Lean Startup* (2011), drawing on his observations across hundreds of startups. The taxonomy reflects Ries's experience at IMVU and his subsequent work as a coach and conference organiser.

WHY IT WAS DEVELOPED

Founders responding to negative evidence either ignored it or pivoted unproductively in random directions. Ries's taxonomy gave specific named directions, each with its own characteristic indicators and risks — converting pivot decisions from intuitive to structured.

FIGURE 1 · TEN PIVOT TYPES

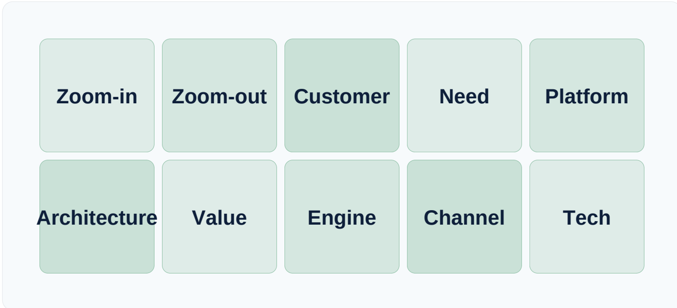

Figure 1. Each pivot type changes specific canvas blocks; pivot is hypothesis-substitution, not goal-abandonment.

HOW IT IS USED

When evidence indicates the current strategy is failing, diagnose which type of pivot the evidence justifies. Each type changes specific BMC blocks. The pivot decision is hypothesis-substitution, not goal-abandonment — preserving venture momentum while changing strategy.

WHEN IT IS USED

Use when validation evidence is consistently negative, in pivot-or-persevere stage-gate reviews, in venture-capital portfolio reviews, and in accelerator-programme weekly check-ins.

BY WHOM IT IS USED

Founders, accelerator coaches, venture-capital partners advising portfolio companies, and corporate-innovation labs deciding whether to continue, pivot or kill internal ventures.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Mintzberg's (1978) emergent-strategy concept named adaptation generally and Christensen's (1997) disruptive-innovation theory described market shifts, Ries's contribution is the specific tactical taxonomy for entrepreneurial pivots. Building on Schumpeter's (1942) creative-destruction tradition, it makes pivot direction a teachable diagnosis.

RELATED FRAMEWORKS IN THIS COURSE

- Build-Measure-Learn · Slibrary 15
- Lemonade · Slibrary 10
- Blank's 4 Steps · Slibrary 16
- Lean Canvas (9 Blocks) · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Ries, E. (2011) *The Lean Startup*. New York: Crown Business.
Maurya, A. (2012) *Running Lean*. 2nd edn. Sebastopol: O'Reilly.
Blank, S. (2013) 'Why the Lean Start-Up Changes Everything', *Harvard Business Review*, 91(5), pp. 63–72.
Mintzberg, H. (1978) 'Patterns in Strategy Formation', *Management Science*, 24(9), pp. 934–948.

Bootstrapping

SLIBRARY 17 · SHEET 1 / 5

Fund the business from revenue and founder savings

PROFESSIONAL DEFINITION

Bootstrapping is the practice of funding venture growth from organic revenue and founder savings rather than external capital. Bootstrapped firms preserve founder ownership, control and discipline at the cost of slower growth and constrained strategic moves. The approach inverts the venture-capital playbook — winning by being capital-efficient rather than by deploying capital faster than competitors. Famous examples include MailChimp, Basecamp, and Atlassian's early years (Greene and Kromer, 2018).

WHO DEVELOPED IT

Long-standing entrepreneurial practice; codified in the modern context by Jason Fried and David Heinemeier Hansson at Basecamp (*Rework*, 2010), and formalised in Greene and Kromer's *The Bootstrapper's Manifesto* (2018). Conceptual roots in Penrose's (1959) growth-of-the-firm theory.

WHY IT WAS DEVELOPED

Venture-capital-backed growth was the dominant playbook of the 2000s–2010s, but most ventures should not raise venture capital — the model demands ten-year exit horizons and 10×+ outcomes that mismatch most businesses. Bootstrapping articulated the alternative path explicitly.

FIGURE 1 · BOOTSTRAP CYCLE

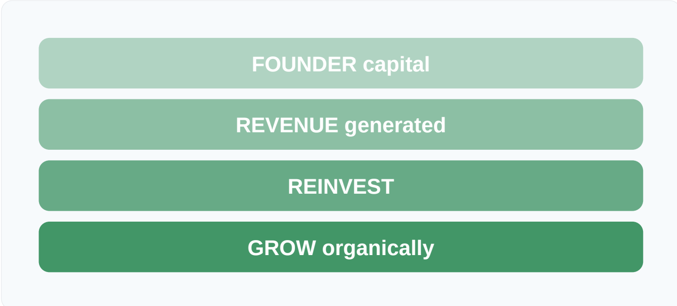

Figure 1. Each cycle preserves ownership and control at the cost of growth speed. The trade-off is the design.

HOW IT IS USED

Constrain spend to current revenue (post early founder investment). Hire only when revenue justifies it. Pursue durable cash-generating customer segments. Treat each month's profit as the funding round.

WHEN IT IS USED

Use when target market is too small for VC-scale outcomes, when founders prioritise control and lifestyle over scale, in capital-light service-business contexts, and in markets where speed is not the limiting factor.

BY WHOM IT IS USED

Founders building lifestyle or sustainable-growth businesses, family-business inheritors growing legacies, second-time founders prioritising independence, and consulting-derivative product businesses.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Schumpeter (1942) emphasised creative destruction through capital-deployed innovation and venture-capital practice (Doriot, 1946) optimised for growth-rate, the modern bootstrapping discourse's contribution is articulating capital-efficiency as a deliberate strategy. Building on Penrose's (1959) growth-of-the-firm tradition, it gives founders a coherent alternative.

RELATED FRAMEWORKS IN THIS COURSE

FFF + Angel · Slibrary 17 | Seed Round · Slibrary 17 | Cap Table Dynamics · Slibrary 17 | Affordable Loss · Slibrary 10

SOURCES (HARVARD REFERENCING STYLE)

Greene, T. and Kromer, S. (2018) *The Bootstrapper's Manifesto*. Self-published.
Fried, J. and Heinemeier Hansson, D. (2010) *Rework*. New York: Crown Business.
Sahlman, W.A. (1990) 'The Structure and Governance of Venture-Capital Organizations', *Journal of Financial Economics*, 27(2), pp. 473–521.
Penrose, E.T. (1959) *The Theory of the Growth of the Firm*. Oxford: Blackwell.

FFF + Angel

SLIBRARY 17 · SHEET 2 / 5

Friends, Family & Fools, then angel investors

PROFESSIONAL DEFINITION

The FFF + Angel funding stage covers the earliest external capital: *Friends, Family & Fools* typically provide \$10K–\$100K based on personal trust at pre-product stage, while *angel investors* deploy \$25K–\$500K based on conviction in founders before traction. Both fund types accept extreme risk in exchange for early valuations; both substitute personal/professional relationships for institutional due diligence. Together they bridge the venture from inception to seed-round eligibility (Wong, Bhatia and Freeman, 2009).

WHO DEVELOPED IT

FFF terminology originated in startup folklore from the 1990s; angel-investing as a structured asset class was professionalised by Bill Payne (*The Definitive Guide to Raising Money from Angels*, 2006) and by groups like Tech Coast Angels and Sand Hill Angels. Wong, Bhatia and Freeman (2009) provided the academic synthesis.

WHY IT WAS DEVELOPED

Pre-traction ventures could not access seed funds (which require evidence) but needed capital to produce that evidence. FFF and angel capital filled the gap with relationship-based diligence — trust in founders rather than in metrics — at sub-institutional ticket sizes.

FIGURE 1 · PRE-SEED TIERS

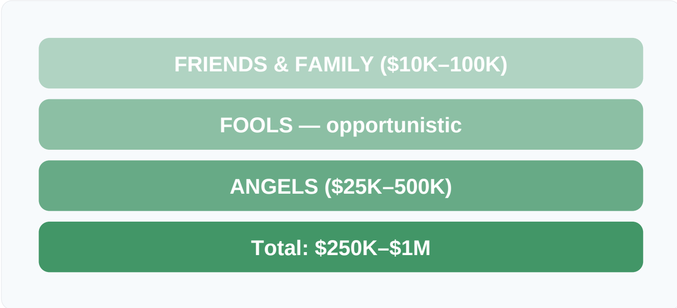

Figure 1. Relationship-based diligence substitutes for institutional metrics in the pre-traction tier.

HOW IT IS USED

Identify potential FFF investors and angels in your network. Structure as SAFE (Y Combinator's standard, 2013) or convertible note to defer valuation. Target \$250K–\$1M total to fund the next 6–12 months until seed-round metrics are achievable.

WHEN IT IS USED

Use after problem-solution fit is demonstrated but before traction sufficient for institutional capital, in pre-seed accelerator graduations, and in serial-founder second-venture launches.

BY WHOM IT IS USED

First-time founders post-discovery, accelerator graduates, and serial entrepreneurs leveraging existing network capital. Standard in technology-startup contexts; rare in capital-intensive sectors.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where venture capital (Doriot, 1946; Sahlman, 1990) operated at \$1M+ ticket sizes, the FFF + Angel layer's contribution is the relationship-based pre-institutional capital tier. Building on Wong et al.'s (2009) academic formalisation, the SAFE instrument (Y Combinator, 2013) reduced legal cost making the layer scalable.

RELATED FRAMEWORKS IN THIS COURSE

Bootstrapping · Slibrary 17 | Seed Round · Slibrary 17 | Series A/B/C · Slibrary 17 | Cap Table Dynamics · Slibrary 17

SOURCES (HARVARD REFERENCING STYLE)

Wong, A., Bhatia, M. and Freeman, Z. (2009) 'Angel Finance: The Other Venture Capital', *Strategic Change*, 18(7–8), pp. 221–230.
Payne, B. (2006) *The Definitive Guide to Raising Money from Angels*. Bill Payne Press.
Sohl, J.E. (2003) 'The Private Equity Market in the USA', *Venture Capital*, 5(1), pp. 29–46.
Y Combinator (2013) *Simple Agreement for Future Equity (SAFE)*. Available at: ycombinator.com.

Seed Round

SLIBRARY 17 · SHEET 3 / 5

$500K–$3M to prove PMF — seed funds, super-angels, accelerator demo day

PROFESSIONAL DEFINITION

The Seed Round is the first institutional venture-capital round, typically $500K–$3M (median ~$2M in 2023), deployed to find product-market fit. Investors are seed-stage venture funds (e.g., Y Combinator Continuity, Sequoia Seed), super-angels, and corporate VC arms. Round structure is usually priced equity at a $5M–$15M post-money valuation; SAFE instruments are common at the lower end. Founders typically take a 12–18 month runway target (NVCA, 2023; Feld and Mendelson, 2022).

WHO DEVELOPED IT

The seed-round category emerged as a distinct stage in the late 2000s as Y Combinator (founded 2005) and similar accelerators created investor demand for very-early ventures. Codified in venture-finance practice by Brad Feld and Jason Mendelson's *Venture Deals* (2022) and the National Venture Capital Association's standard documents.

WHY IT WAS DEVELOPED

The gap between angel capital (sub-$500K) and Series A (post-PMF $5M+) left a funding void during the period when ventures most needed capital. Seed funds emerged to fill it, with diligence calibrated to pre-PMF metrics rather than the revenue traction Series A required.

FIGURE 1 · SEED ROUND ANATOMY

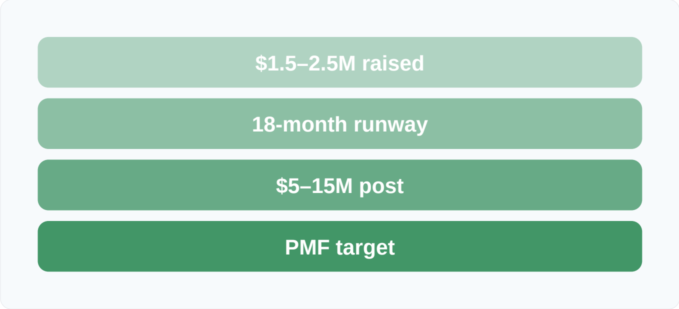

Figure 1. The seed round funds the search for PMF; if PMF is not achieved, Series A is unavailable.

HOW IT IS USED

Target $1.5–$2.5M against a 18-month runway plan. Pitch seed-stage VCs and super-angels. Use SAFE or priced-equity round depending on valuation range. Reserve 10–15% option pool for hires. Close in 6–12 weeks.

WHEN IT IS USED

Use after FFF/angel capital is exhausted but before traction sufficient for Series A, after problem-solution fit but before product-market fit, and as the funding-stage during which PMF must be achieved.

BY WHOM IT IS USED

Post-accelerator founders, second-time founders with track record, and corporate-innovation spinouts. Standard for technology, B2B SaaS, consumer-internet and biotech ventures.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Doriot's (1946) original ARDC venture capital deployed at industrial-scale and Series A funding required revenue traction, the seed round's contribution is the institutionalisation of pre-PMF capital. Building on Y Combinator's (2005) accelerator-to-investor pipeline, the seed category became standard by 2015.

RELATED FRAMEWORKS IN THIS COURSE

FFF + Angel · Slibrary 17
Series A/B/C · Slibrary 17
Cap Table Dynamics · Slibrary 17
Product-Market Fit · Slibrary 14

SOURCES (HARVARD REFERENCING STYLE)

Feld, B. and Mendelson, J. (2022) *Venture Deals*. 5th edn. Hoboken: Wiley.
National Venture Capital Association (2023) *NVCA Yearbook*. Washington, DC: NVCA.
Kupor, S. (2019) *Secrets of Sand Hill Road*. New York: Portfolio.
Sahlman, W.A. (1990) 'The Structure and Governance of Venture-Capital Organizations', *Journal of Financial Economics*, 27(2), pp. 473–521.

Series A/B/C

SLIBRARY 17 · SHEET 4 / 5

A: scale GTM. B: scale ops. C: expand market — each round has a different bar

PROFESSIONAL DEFINITION

Series A, B and C are the three main scale-stage venture-capital rounds, each with a different financial profile and milestone gate. *Series A* ($5–15M, post-PMF) funds go-to-market scale-up. *Series B* ($15–50M, post-repeatable-GTM) funds operational scaling — international, hiring, infrastructure. *Series C+* ($50M+, post-scaling) funds market expansion or pre-IPO consolidation. Each round's diligence bar reflects the stage's specific risk (Feld and Mendelson, 2022; Kupor, 2019).

WHO DEVELOPED IT

Stage-named rounds emerged from US venture-capital practice in the 1980s. The current naming and milestone-bar conventions were standardised by the National Venture Capital Association (NVCA) and codified for practitioners by Feld and Mendelson's *Venture Deals* (2022) and Kupor's *Secrets of Sand Hill Road* (2019).

WHY IT WAS DEVELOPED

Lumping all post-seed rounds together obscured what each was actually for. The A/B/C/D distinction (and now D, E, F+ at growth stages) provides shared vocabulary for matching capital to specific milestone-driven needs and informing investor and founder expectations.

FIGURE 1 · THREE STAGES

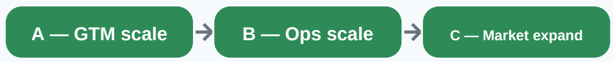

Figure 1. Each round's bar reflects specific stage risk; A funds GTM, B funds ops, C funds expansion.

HOW IT IS USED

Series A: post-PMF, scale GTM, target $1M+ ARR. Series B: post-repeatable-GTM, scale ops, target $5–10M ARR. Series C: post-scaling, expand market, target $20M+ ARR. Each round's diligence focuses on the stage's specific risk.

WHEN IT IS USED

Use as the funding-stage roadmap for venture-capital-backed scaling ventures. Each round's timing depends on milestone achievement, not calendar.

BY WHOM IT IS USED

Founders of post-PMF venture-backed companies, venture-capital partners across the stages, growth-stage CFOs, and corporate-development teams considering acquiring or partnering with portfolio companies.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Doriot's (1946) original ARDC made one-shot growth investments and 1980s VC operated without standardised stage names, the modern A/B/C convention's contribution is the milestone-stage matching that aligns capital sizes to specific risk profiles. Building on agency-theory governance research (Sahlman, 1990), it gives stage-funding its standard architecture.

RELATED FRAMEWORKS IN THIS COURSE

- Seed Round · Slibrary 17
- Cap Table Dynamics · Slibrary 17
- Product-Market Fit · Slibrary 14
- AARRR · Pirate Metrics · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Feld, B. and Mendelson, J. (2022) *Venture Deals*. 5th edn. Hoboken: Wiley.
Kupor, S. (2019) *Secrets of Sand Hill Road*. New York: Portfolio.
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National Venture Capital Association (2023) *NVCA Yearbook*. Washington, DC: NVCA.

Cap Table Dynamics

SLIBRARY 17 · SHEET 5 / 5

Founders · Employees · Seed · Series A — dilution compounds

PROFESSIONAL DEFINITION

Cap Table Dynamics describes how ownership percentages evolve across funding rounds. Each round dilutes existing holders proportionally; option-pool top-ups dilute founders specifically. After typical Seed → A → B → C progression, founder ownership commonly drops from 100% to 15–25%, and ownership ≠ control because preferred-stock terms (liquidation preferences, board seats, protective provisions) shift control to investors before equity dilution alone would. Modelling cap-table evolution is essential pre-fundraising work (Feld and Mendelson, 2022).

WHO DEVELOPED IT

Cap-table modelling practice emerged from venture-capital legal and finance practice in the 1990s; standardised in modern form through Feld and Mendelson's *Venture Deals* (2022), Kupor's *Secrets of Sand Hill Road* (2019), and tools like Carta and Pulley.

WHY IT WAS DEVELOPED

Founders raised rounds without modelling future dilution, then woke up post-Series-C with single-digit ownership. Cap-table modelling tools and the educational literature emerged to make dilution explicit before, not after, signing — the most important founder-protection literacy.

FIGURE 1 · DILUTION EVOLUTION

Round	Founder %
Founding	100%
Seed	70%
Series A	50%
Series B	30%
Series C	20%

Figure 1. Typical founder dilution path; ownership ≠ control because preferred-stock terms shift control earlier.

HOW IT IS USED

Build a cap table model with each round's dilution, option-pool top-ups, and exit waterfall (preferences applied at sale). Evaluate each fundraising decision against multiple-round projection, not single-round optics.

WHEN IT IS USED

Use before any fundraising round, in option-pool sizing, in co-founder equity-split conversations, in M&A; waterfall analysis, and in employee-equity communication.

BY WHOM IT IS USED

Founders, CFOs, startup lawyers, venture-capital partners, employees considering equity-vs-cash trade-offs, and acquirers modelling exit-waterfall implications.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where traditional shareholder-equity analysis (Modigliani and Miller, 1958) treated capital structure abstractly, cap-table modelling's contribution is the multi-round dilution-and-control projection at venture stage. Building on Sahlman's (1990) governance-of-VC research, the modern modelling tradition makes founder economics auditable round-by-round.

RELATED FRAMEWORKS IN THIS COURSE

Seed Round · Slibrary 17 | Series A/B/C · Slibrary 17 | FFF + Angel · Slibrary 17 | Internal Term Sheet · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

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Kupor, S. (2019) *Secrets of Sand Hill Road*. New York: Portfolio.
Sahlman, W.A. (1990) 'The Structure and Governance of Venture-Capital Organizations', *Journal of Financial Economics*, 27(2), pp. 473–521.
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CAC

SLIBRARY 18 · SHEET 1 / 5

Customer Acquisition Cost = total S&M ÷ new customers

PROFESSIONAL DEFINITION

Customer Acquisition Cost (CAC) is total sales and marketing spend over a period divided by the number of new customers acquired in that period. CAC is the foundational unit-economics metric: it answers *how much does it cost to win a customer?* and combines with LTV to determine whether a business model is viable. Blended CAC (across all channels) is the headline metric; channel-specific and segment-specific CAC inform allocation decisions (Skok, 2017; Croll and Yoskovitz, 2013).

WHO DEVELOPED IT

CAC as a measured metric emerged from direct-marketing practice in the 1980s; codified for software-as-a-service contexts by David Skok (Matrix Partners) in his "SaaS Metrics 2.0" framework (2010, updated 2017) and operationalised in *Lean Analytics* (Croll and Yoskovitz, 2013).

WHY IT WAS DEVELOPED

SaaS businesses with negative early-stage cash flow needed a way to demonstrate underlying economics worked even while losing money. CAC made that demonstrability concrete: if CAC is much lower than LTV and payback is reasonable, the negative cash flow is investment, not loss.

FIGURE 1 · CAC COMPOSITION

Component	Spend
Paid ads	$120K
Sales team	$80K
Content & SEO	$40K
Customers won	60 new

Figure 1. CAC = $240K ÷ 60 = $4,000. Fully-loaded calculation includes all S&M; overhead.

HOW IT IS USED

Calculate fully-loaded CAC = (sales + marketing total spend) ÷ new customers acquired in the same period. Track blended and per-channel. Segment by customer profile to surface where CAC is highest and lowest.

WHEN IT IS USED

Use as a continuous monthly metric, in fundraising-pitch unit-economics tables, in growth-channel allocation decisions, and in board-level efficiency-of-growth diagnostics.

BY WHOM IT IS USED

Growth marketers, CFOs, founders pitching investors, venture-capital portfolio analysts, and corporate-development teams evaluating acquisition targets.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where direct-marketing's cost-per-response (Hopkins, 1923) measured campaign-level acquisition cost, Skok's contribution is fully-loaded SaaS-context CAC including all sales-and-marketing overhead. Building on Reichheld's (1996) loyalty-economics tradition, it became the SaaS unit-economics anchor.

RELATED FRAMEWORKS IN THIS COURSE

LTV · Slibrary 18 | LTV/CAC Ratio · Slibrary 18 | Payback Period · Slibrary 18 | AARRR · Pirate Metrics · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Skok, D. (2017) 'SaaS Metrics 2.0', *For Entrepreneurs blog*.
Croll, A. and Yoskovitz, B. (2013) *Lean Analytics*. Sebastopol: O'Reilly.
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LTV

SLIBRARY 18 · SHEET 2 / 5

Customer Lifetime Value = ARPU × gross margin × lifetime

PROFESSIONAL DEFINITION

Customer Lifetime Value (LTV) is the total gross profit a customer generates over their entire relationship with the firm, calculated as ARPU (average revenue per user) × gross margin × average customer lifetime (1 ÷ churn rate). LTV pairs with CAC to test viability: LTV/CAC > 3× is the SaaS rule of thumb. Sub-1× LTV/CAC means losing money on every customer — a model that gets worse with growth, not better (Skok, 2017; Reichheld, 1996).

WHO DEVELOPED IT

LTV originated in direct-marketing literature (Reichheld's *The Loyalty Effect*, 1996) and was operationalised for SaaS by David Skok (2010, 2017). The mathematical form (cohort-based) was refined by SaaS-finance practitioners like Tomasz Tunguz and Jason Lemkin through the 2010s.

WHY IT WAS DEVELOPED

Single-period revenue analysis missed that valuable customers were those who stayed and grew, not just those who paid initially. LTV captured that durability quantitatively — making it possible to invest more in acquiring valuable customers than in acquiring transactional ones.

FIGURE 1 · LTV COMPONENTS

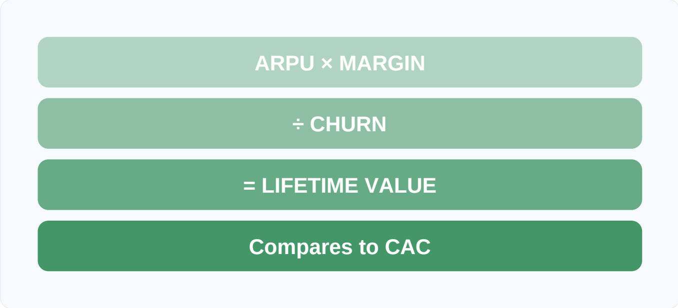

Figure 1. LTV captures retention's compounding value; pure first-period revenue understates durable customer worth.

HOW IT IS USED

Calculate LTV = ARPU × gross margin% × (1 ÷ monthly churn). For example: $100 × 75% × (1 ÷ 0.05) = $1,500. Recalculate periodically as churn improves; segment LTV by customer profile to inform CAC-allocation decisions.

WHEN IT IS USED

Use as a continuous metric, in fundraising-pitch unit-economics tables, in customer-segment investment decisions, and in product-roadmap prioritisation (features that reduce churn raise LTV).

BY WHOM IT IS USED

Growth marketers, CFOs, founders, customer-success leaders, and venture-capital partners diagnosing portfolio-company unit-economics health.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Reichheld's (1996) loyalty-economics work showed retention's compounding value qualitatively, Skok's contribution is the standardised SaaS-context LTV formula linking ARPU, margin and churn. Building on Bayes-rule cohort analysis traditions, it gave SaaS unit economics its standard right-hand side.

RELATED FRAMEWORKS IN THIS COURSE

- CAC · Slibrary 18
- LTV/CAC Ratio · Slibrary 18
- Payback Period · Slibrary 18
- AARRR · Pirate Metrics · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Skok, D. (2017) 'SaaS Metrics 2.0', *For Entrepreneurs blog*.
Reichheld, F.F. (1996) *The Loyalty Effect*. Boston: Harvard Business School Press.
Croll, A. and Yoskovitz, B. (2013) *Lean Analytics*. Sebastopol: O'Reilly.
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LTV/CAC Ratio

SLIBRARY 18 · SHEET 3 / 5

Rule of thumb: ≥ 3 — below 1 means losing money on every customer

PROFESSIONAL DEFINITION

The LTV/CAC ratio is the central unit-economics health indicator: it compares customer lifetime value to customer acquisition cost. The widely-cited SaaS rule of thumb says ratio ≥ 3× indicates a sustainable model, between 1× and 3× indicates a marginal model needing optimisation, and below 1× indicates losing money on every customer. The ratio sets the upper bound on capital deployment efficiency for a venture (Skok, 2017; Bessemer, 2020).

WHO DEVELOPED IT

The LTV/CAC ratio framing was operationalised by David Skok (Matrix Partners) in his SaaS Metrics 2.0 work (2010 onwards), with Bessemer Venture Partners' "State of the Cloud" reports (2014–present) institutionalising the 3× benchmark for the SaaS investor community.

WHY IT WAS DEVELOPED

Single-metric analyses (CAC alone or LTV alone) missed the relationship that mattered. Skok's ratio framing made the joint test the headline metric — a single number that captured whether the unit economics were sustainable or not.

FIGURE 1 · RATIO BANDS

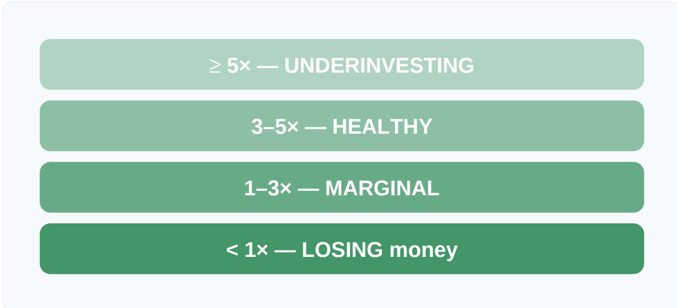

Figure 1. The 3× threshold is the SaaS standard; very high ratios may indicate growth-underinvestment.

HOW IT IS USED

Calculate LTV and CAC separately, take their ratio. Below 1×: stop acquiring (each customer is a loss). 1–3×: optimise both sides. Above 3×: invest more in acquisition. Above 5× often indicates underinvestment in growth, not virtue.

WHEN IT IS USED

Use in monthly board reports, in fundraising-pitch unit-economics tables, in growth-investment decisions, and in venture-capital portfolio efficiency diagnostics.

BY WHOM IT IS USED

Growth marketers, CFOs, founders pitching investors, venture-capital partners, and corporate-development teams evaluating SaaS acquisitions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where individual CAC (Hopkins, 1923) and LTV (Reichheld, 1996) traditions evolved separately, Skok's contribution is their ratio-as-headline-metric. Building on Reichheld's (1996) loyalty-economics framing, it gives SaaS unit-economics a single auditable number for board-level discussion.

RELATED FRAMEWORKS IN THIS COURSE

CAC · Slibrary 18 | LTV · Slibrary 18 | Payback Period · Slibrary 18 | AARRR · Pirate Metrics · Slibrary 16

SOURCES (HARVARD REFERENCING STYLE)

Skok, D. (2017) 'SaaS Metrics 2.0', *For Entrepreneurs blog*.
Bessemer Venture Partners (2020) *State of the Cloud Report*. Available at: bvp.com.
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Payback Period

SLIBRARY 18 · SHEET 4 / 5

Months of gross margin to earn back CAC — SaaS benchmark < 12 months

PROFESSIONAL DEFINITION

CAC Payback Period is the number of months of gross margin required to earn back the acquisition cost: CAC ÷ (monthly ARPU × gross margin%). A 12-month benchmark is standard for SaaS (Bessemer, 2020); shorter is healthier. The metric matters because it determines cash-flow timing — even with good LTV/CAC, long payback periods consume capital faster than fast-payback models with the same ratio. Cash-constrained ventures need short payback (Skok, 2017).

WHO DEVELOPED IT

Payback-period metric originated in capital-budgeting practice (1950s); applied to customer-acquisition-cost framing by David Skok in SaaS Metrics 2.0 (2010 onwards) and standardised into the SaaS investor community by Bessemer Venture Partners' annual reports.

WHY IT WAS DEVELOPED

LTV/CAC alone said the model worked but said nothing about cash-flow timing. Two ventures with identical 4× LTV/CAC ratios could have radically different capital needs depending on how long it takes to recover CAC — payback period made that timing explicit.

FIGURE 1 · PAYBACK BANDS

Figure 1. Payback period determines cash-flow timing; long payback consumes capital faster than fast-payback at identical LTV/CAC.

HOW IT IS USED

Calculate CAC ÷ (ARPU × gross margin%). For example: \$4,000 ÷ (\$100 × 75%) = 53 months — a long payback indicating the model needs significant working capital. Compare to <12 month benchmark for SaaS efficiency.

WHEN IT IS USED

Use in cash-runway planning, in fundraising-pitch unit-economics tables, in growth-investment timing decisions, and in venture-capital portfolio cash-flow diagnostics.

BY WHOM IT IS USED

CFOs, founders managing cash runway, growth marketers, venture-capital partners assessing capital-efficiency, and corporate-development teams modelling acquisition cash-flow impact.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where capital-budgeting payback period (Hertz, 1964) was applied to fixed-asset investments, Skok's contribution is the customer-acquisition application that surfaces unit-economics cash-timing. Building on Reichheld's (1996) loyalty-economics tradition, it complements LTV/CAC ratio with the missing time dimension.

RELATED FRAMEWORKS IN THIS COURSE

CAC · Slibrary 18 | LTV · Slibrary 18 | LTV/CAC Ratio · Slibrary 18 | Contribution Margin & Burn · Slibrary 18

SOURCES (HARVARD REFERENCING STYLE)

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Contribution Margin & Burn

SLIBRARY 18 · SHEET 5 / 5

CM × volume – fixed costs = profit; burn = fixed – contribution

PROFESSIONAL DEFINITION

Contribution Margin (CM) is per-unit revenue minus per-unit variable cost — the dollars each additional customer contributes toward fixed costs and profit. Burn is monthly negative cash flow when contribution margin × volume is less than fixed costs (R&D;, G&A;, sales-overhead). The two metrics together show when a venture transitions from burn to profit: when CM × volume exceeds fixed costs, the firm is profitable; until then, capital fills the gap (Mohnen and Bourgault, 2003; Croll and Yoskovitz, 2013).

WHO DEVELOPED IT

Contribution margin originated in cost-accounting practice (Goldratt, 1984; Horngren, 1972). Burn-rate framing emerged from venture-capital practice in the 1990s and was operationalised for startups in *Lean Analytics* (Croll and Yoskovitz, 2013) and Maurya's *Scaling Lean* (2016).

WHY IT WAS DEVELOPED

Founders confused gross margin (revenue minus COGS) with contribution margin (revenue minus all variable costs). The distinction matters because gross margin can look healthy while contribution margin is negative — and contribution margin is what determines whether scale produces profit.

FIGURE 1 · BURN → PROFIT

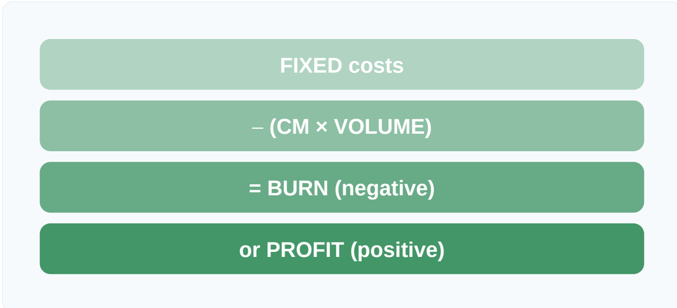

Figure 1. The threshold volume = fixed ÷ CM. Until that volume, capital fills the gap; after, the firm self-funds.

HOW IT IS USED

Calculate per-unit contribution margin (revenue minus all variable costs). Calculate fixed costs (R&D;, G&A;, sales overhead). Burn = fixed – (CM × volume). Profitability threshold = fixed ÷ CM. Plan capital runway against the gap.

WHEN IT IS USED

Use in monthly board reports, in cash-runway planning, in pricing-strategy reviews, and in growth-investment timing decisions where the margin/scale relationship determines viability.

BY WHOM IT IS USED

CFOs, founders managing cash runway, financial planning and analysis (FP&A;) teams, venture-capital portfolio analysts, and corporate-development teams modelling acquisition financial impact.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Goldratt's (1984) Theory of Constraints emphasised throughput accounting and Horngren's (1972) management-accounting tradition framed contribution-margin reporting, the SaaS-startup synthesis's contribution is the burn-vs-contribution model linking unit economics to cash runway. Building on Sahlman's (1990) venture-finance research, it gives founders the cash-survival diagnostic.

RELATED FRAMEWORKS IN THIS COURSE

LTV/CAC Ratio · Slibrary 18 | Payback Period · Slibrary 18 | CAC · Slibrary 18 | Cap Table Dynamics · Slibrary 17

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